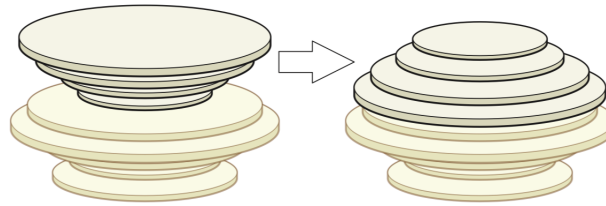


1. (20%) Please answer the following questions:
 - (a) Illustrate the operation of RADIX-SORT on the following numbers: 457, 654, 578, 254, 957, 499, 558, 911, 731, 536.
 - (b) In the algorithm SELECT, the input elements are divided into groups of 5. Will the algorithm work in linear time if they are divided into groups of 7?
 - (c) Demonstrate the insertion of keys 36, 27, 15, 47, 82, 33 into a hash table with collisions resolved by chaining. Let the table have 5 slots, and let the hash function be $h(k) = k \bmod 5$.
2. (20%) A red-black tree is a balanced binary tree that has height $O(\log n)$.
 - (a) Show that a red-black tree with n internal nodes has height $\leq 2 \lg(n + 1)$.
 - (b) Show the red-black trees that result after successively inserting the keys 36, 27, 15, 47, 82, 33 into an initially empty red-black tree.
3. (20%) Given a sequence $X = \langle x_1, x_2, \dots, x_m \rangle$, the longest increasing subsequence problem is to find an increasing subsequence of X with the longest length.
 - (a) Find a longest increasing subsequence of $\langle 36, 27, 15, 47, 82, 33, 19 \rangle$.
 - (b) Describe an algorithm that solves the longest increasing subsequence problem in $O(n^2)$ time.
4. (20%) Search trees are data structures that support many dynamic-set operations. Dynamic-set operations includes SEARCH, MINIMUM, MAXIMUM, PREDECESSOR, SUCCESSOR, INSERT, and DELETE.
 - (a) Show that if a node in a binary search tree has two children, then its successor has no left child and its predecessor has no right child.
 - (b) For the set of keys $\{36, 27, 15, 47, 82, 33, 19\}$, draw binary search trees of height 2, 3, 4, 5 and 6.
 - (c) Write recursive versions of the SEARCH and TREE-MAXIMUM procedures.
5. (15%) In this question, you are given a set $C = \{1, 5, 10, 25\}$ of coin denominations, that is, pennies, nickels, dimes, and quarters. Your goal is to design an algorithm which, for any input value V in cents, outputs the minimum number of coins with denominations in C whose total value is exactly V . For example, to represent the value $V = 42$, we would choose one quarter, one dime, one nickel, and two pennies, a total of 5 coins. Your algorithm should take $O(|C|) = O(1)$ time.
 - (a) Describe your algorithm, argue briefly that it does indeed achieve the desired running time.
 - (b) Prove formally that it outputs the minimum number of coins needed to represent a given value V , for any value V it is given as input.

6. (15%) Suppose you are given a stack of n pancakes of different sizes. You want to sort the pancakes so that smaller pancakes are on top of larger pancakes. The only operation you can perform is a flip—insert a spatula under the top of k pancakes, for some integer k between 1 and n , and flip them all over.



- (a) Describe an algorithm to sort an arbitrary stack of n pancakes using as few flips as possible.
- (b) Exactly how many flips does your algorithm perform in the worst case?